STAT 5200, Notes 1[1]

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Readings. Wooldridge (W). Read Chapters 1 and 2. Chapter 2 provides important foundation. Section 2.1 gives definitions and perspective. Section 2.2.1 introduces conditional expectation and gives examples. Section 2.2.2 describes partial effects and elasticities. In Section 2.2.3 an error term is added to the conditional expectation to obtain the response variable. The error term was included in the discussion in Chapter 1. Some properties of conditional expectations are discussed in Section 2.2.4. Section 2.2.5 gives a detailed discussion of average partial effects, to take into account the impact of unobserved variables. Linear projection is fundamental to understanding the development of theory and methodology in W, and it is treated in Section 2.3, where the difference between conditional expectation and linear projection is explained. Appendix 2A gives properties of conditional expectations and conditional variances (in 2.A.1 and 2.A.2, respectively), and properties of linear projections (in 2.A.3).

This is important material, and you may find parts of it heavy going, especially if much of it is new to you. The material in Chapter 2 gives you a good foundation for moving forward. As we progress, I will revisit some of this material as we encounter methodology and theory. Be patient and give yourself time to become comfortable with the concepts and the mathematics.

Causal Relationships and the Random Sampling Assumption. Historically the statistics literature has focused on randomized experimentation. In one example, the researcher decides which subjects will receive one treatment and which will receive a second treatment, assigning subjects at random to the two groups. Then the two group responses are compared, and the calculated difference assesses causality. This approach has its origin in agricultural and industrial experimentation, where the subjects are passive units (such as plots of land, or production batches). By contrast, the economist has worked in settings where the subjects themselves make decisions about treatment assignment. For example, individual subjects decide whether or not to enroll

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for additional education. For economists the subjects are individuals, families, firms, etc. The statistician has benefited from random assignment of subjects to treatments in seeking to determine causality. The economist has faced more obstacles in addressing causality and has had to develop additional tools and methods.

There are several issues which historically have distinguished the practice of econometrics from that of statistics.

(i) One is the emphasis upon addressing causal effects, as opposed to simply determining that variables are associated (correlated). That is, one wants to know whether a change in a particular explanatory variable *causes* a change in the response variable. To do so, one needs to establish controls on the other relevant explanatory variables. Often, however, some of these other explanatory variables cannot be observed, or they are observed with measurement error; when this is the case, one can often still make headway, by introducing proxy variables or instrumental variables, for example, methods developed largely by economists.

In recent years, though, statisticians have been addressing experiments without randomization and have been increasingly paying attention to causality. For both economists and statisticians, subjects who do not comply with the assigned treatment protocol, or who fail to provide data at assigned measurement times, or who leave the study before it is completed, pose substantial difficulties in assessing causality.

Here are some examples of questions which deal with causality and problems in estimation:

1. Consider a group of elderly persons who go to hospital emergency rooms for a particular problem. Some receive treatment and are sent home, while others are hospitalized. What is the effectiveness of the hospitalization?

2. For persons in the labor force, what is the effect of an added year of education upon earnings? That is, what is the return to education?

3. How will an increase of the minimum wage by one dollar affect the unemployment rate? How will it affect the prices of consumer goods and services? How will it affect the amount of government assistance to the poor?

4. What is the predicted effect upon life expectancy of a ten percent increase in GDP per capita?

5. What is the estimated effect of cigarette smoking during pregnancy upon the weight of newborns?

6. Many organizations, including University of Pennsylvania, sponsor wellness programs for their employees. The aims of these programs are to improve health and lower the cost of health

insurance borne by the organization and the employees. Participation is voluntary. Do these programs succeed in improving health outcomes and lowering costs?

7. A study published 25 July 2017 in *Journal of the American Medical Association* reported finding signs of chronic traumatic encephalopathy (CTE) in the brains of 177 out of 202 deceased football players. The brains were donated for study by their families. The men had played football in high school, in college, in semiprofessional settings, in the Canadian Football League, and in the National Football League. In the last two groups, 7 out of 8 and 110 out of 111 cases revealed evidence of CTE. Can one conclude there is a causal link between playing football and CTE?

8. During the coronavirus pandemic, attention focused on the reproduction number R_t, which assesses the transmission rate of the virus. It is defined as the average number of people who become infected by an infectious person. Note that it is a function of time. How can the reproduction number be accurately estimated?

(ii) A second feature is the use in econometrics of a collection of distinctive statistical tools. Perhaps the most fundamental of these is the instrumental variable (IV). IVs allow one to tackle estimation bias caused by unobserved missing variables in a regression, e.g. In addition, econometricians use a wide array of tests typically not encountered in the work of statisticians.

(iii) Econometricians use a random sampling assumption. That is, there is some population of interest, and, in the simplest such framework, an independent and identically distributed (i.i.d.) random sample is drawn from this population. This assumes that the explanatory variable data, as well as the response variable data, are random selections from a population. This is in contrast to the classical linear model considered in statistics, where the explanatory variable data are treated as fixed values and distributional assumptions are made about the error term. The assumption of fixed explanatory variable data makes sense for designed experiments, such as in agricultural and industrial settings, which played a prominent role in the historical development of statistical methodology.

There are considerable advantages and good reason to work with the random sampling assumption. In econometric settings it is typically not realistic to treat the explanatory variable data as fixed values. Moreover, with the random sampling assumption, there is no need to use matrix notation and to consider properties of data matrices. Parameter estimates, though, are biased in finite samples, as opposed to the classical statistics setting with fixed explanatory variables. Also, if cross sections are observed at different points in time, the i.i.d. sampling assumption needs to be relaxed, e.g., to independent, nonidentically distributed (inid) sampling. Moreover, some other methods of sampling, such as cluster sampling and stratified sampling, are sometimes employed, and these also need to be addressed. Further, some types of sampling lead to time series or panel data models, which incorporate spatial and/or temporal correlation.

The discussion in Section 1.4 of W explains why, in many econometric settings, it is not appropriate to assume the explanatory variables are nonrandom. A consequence of the random sampling assumption is that one must focus on asymptotic properties of estimators, rather than their finite sample properties. The asymptotics operate as the dimension N of the cross section tends to infinity. If there is also a time series dimension T, in econometric settings it will typically remain fixed and will be a relatively small number.

For most estimators we will consider, the finite sample properties of estimators and procedures are too difficult to describe analytically, and hence we focus on asymptotics.

Conditional Expectation. Let y be a random response variable and $\mathbf{x} = (x_1, ..., x_K)$ a $1 \times K$ random vector of explanatory variables. We will be concerned with inference for parametric models of the *conditional expectation* $E[y|\mathbf{x}]$. We adopt the point of view that the conditional expectation defines a *structural* model. Often we will modify a structural model to obtain an estimable model.

The conditional expectation relates the average value of y to $\mathbf{x}$. It is random variable, a function of $\mathbf{x}$.

Conditional Expectation – Definition. Let's do a simple calculation as an example.

I'll use lower case letters for the random variables, but will use upper case subscripts to label the densities. Let y and x be continuous random variables with joint density $f_{Y,X}(y, x)$ and marginal densities $f_Y(y)$ and $f_X(x)$.

The conditional density of y, given $x = \tilde{x}$, is $f_{Y|X}(y|x = \tilde{x}) = f_{Y,X}(y, x = \tilde{x})/f_X(x = \tilde{x})$, and

$$E[y|x] = \int y f_{Y|X}(y|x = \tilde{x})dy.$$

Note that if y and x are independent, the conditional density $f_{Y|X}(y|x)$ is equal to the marginal density $f_Y(y)$, and thus the conditional expectation is equal to $E[y]$. If the random variables are discrete, summations are used in place of integrals, of course, and mixed variables can also be considered.

Properties of Conditional Expectations and Conditional Variances. It is important to understand that the conditional expectation of y given $\mathbf{x}$ is a random variable – it is a function of $\mathbf{x}$.

Properties of the conditional expectation operation are listed and discussed by Wooldridge in Appendix 2.A.1. The simplest property of conditional expectation is an iterated expectation result (Property CE.2),

$$E[y] = E[E[y|\mathbf{x}]].$$

Suppose $\mathbf{x}$ is a random vector that is a function of a random vector $\mathbf{w}$, $\mathbf{x} = \mathbf{f}(\mathbf{w})$, say As Wooldridge remarks at (2.19), a general statement of iterated expectations is

$$E[y|\mathbf{x}] = E\left[E[y|\mathbf{w}] \mid \mathbf{x}\right]. \qquad ((2.19) \text{ in W})$$

This is Property CE.3. At (2.20) Wooldridge notes that, under these given assumptions, it is also true that

$$E[y|\mathbf{x}] = E\left[E[y|\mathbf{x}] \mid \mathbf{w}\right]. \qquad ((2.20) \text{ in W})$$

Wooldridge uses both of these extensively. To help you understand conditional expectation results, remember that the smaller information set always appears in the end expression.

Property CE.1 in W is fundamentally important; it states that the conditional expectation is a linear operator. Linearity follows simply from properties of integrals and summations. I have listed CE.2 and CE.3 above.

Property CE.5 states: If $(\mathbf{u}, \mathbf{v})$ is independent of $\mathbf{x}$, then $E[\mathbf{u}|\mathbf{x}, \mathbf{v}] = E[\mathbf{u}|\mathbf{v}]$. Let's derive this, assuming continuous distributions. The conditional density of $\mathbf{u}$, given $(\mathbf{x}, \mathbf{v})$, is

$$f_{\mathbf{U}|\mathbf{X},\mathbf{V}}(\mathbf{u} \mid \mathbf{x}, \mathbf{v}) = \frac{f_{\mathbf{U},\mathbf{X},\mathbf{V}}(\mathbf{u}, \mathbf{x}, \mathbf{v})}{f_{\mathbf{X},\mathbf{V}}(\mathbf{x}, \mathbf{v})} = \frac{f_{\mathbf{U},\mathbf{V}}(\mathbf{u}, \mathbf{v}) f_{\mathbf{X}}(\mathbf{x})}{f_{\mathbf{V}}(\mathbf{v}) f_{\mathbf{X}}(\mathbf{x})} = \frac{f_{\mathbf{U},\mathbf{V}}(\mathbf{u}, \mathbf{v})}{f_{\mathbf{V}}(\mathbf{v})} = f_{\mathbf{U}|\mathbf{V}}(\mathbf{u} \mid \mathbf{v}),$$

by the stated independence. Then

$$E[\mathbf{u} \mid \mathbf{x}, \mathbf{v}] = \int \mathbf{u} f_{\mathbf{U}|\mathbf{X},\mathbf{V}}(\mathbf{u} \mid \mathbf{x}, \mathbf{v}) d\mathbf{u} = \int \mathbf{u} f_{\mathbf{U}|\mathbf{V}}(\mathbf{u} \mid \mathbf{v}) d\mathbf{u} = E[\mathbf{u} \mid \mathbf{v}].$$

Property CE.6 relates to the error form of a model and is crucial for understanding what we will do in this course. Define the random variable u to be $y - E[y \mid \mathbf{x}]$. Then if $\mathbf{g}(\mathbf{x}) = (g_1(\mathbf{x}), ..., g_J(\mathbf{x}))$ if *any* function satisfying $E|g_j(\mathbf{x})u| < \infty$, $j = 1, ..., J$, and if $E|u| < \infty$, it follows that

$$E[\mathbf{g}(\mathbf{x})u] = \mathbf{0}.$$

Special cases of this result are $E[u] = 0$ and $Cov(x_j, u) = 0$, $j = 1, ..., J$. That is, the error term of a model is orthogonal to the space spanned by the explanatory variables. It's useful to step through the proof of CE.6. First,

$$\begin{aligned} E[u \mid \mathbf{x}] &= E\left[y - E[y \mid \mathbf{x}] \mid \mathbf{x}\right] = E[y \mid \mathbf{x}] - E\left[E[y \mid \mathbf{x}] \mid \mathbf{x}\right] \\ &= E[y \mid \mathbf{x}] - E[y \mid \mathbf{x}] E[1 \mid \mathbf{x}] \\ &= E[y \mid \mathbf{x}] - E[y \mid \mathbf{x}] = 0. \end{aligned}$$

Then

$$E[\mathbf{g}(\mathbf{x})u] = E\left[E[\mathbf{g}(\mathbf{x})u \mid \mathbf{x}]\right] = E\left[\mathbf{g}(\mathbf{x})E[u \mid \mathbf{x}]\right] = E[\mathbf{g}(\mathbf{x}) \cdot 0] = \mathbf{0}.$$

Property CE.7 is Jensen's inequality for conditional expectations. It is needed for Property CE.8.

Property CE.8 uses the second moment of the random variable y, $E[y^2]$, assuming this is finite. The property states that the conditional expectation $E[y|\mathbf{x}]$ is the best mean square predictor of y based on the information content of $\mathbf{x}$, that is, $E[y|\mathbf{x}]$ is the choice of $m(\mathbf{x})$ which minimizes

$$E\left[(y - m(\mathbf{x}))^2\right]$$

among all functions satisfying $Em(\mathbf{x})^2 < \infty$. Thus, the best mean square predictor and the conditional expectation are one and the same.

In Appendix 2.A.2 Wooldridge defines the conditional variance of y given $\mathbf{x}$ and gives some of its properties. The conditional variance is also a random variable. It is defined as

$$Var(y|\mathbf{x}) = E\left[(y - E[y|\mathbf{x}])^2 \mid \mathbf{x}\right] = E\left[y^2|\mathbf{x}\right] - (E[y|\mathbf{x}])^2 .$$

In the context of the conditional distribution of y given $\mathbf{x}$, Property CV.2 states that the variance of y is equal to the expectation of the conditional variance plus the variance of the conditional expectation,

$$Var(y) = E\left[Var(y|\mathbf{x})\right] + Var\left(E[y|\mathbf{x}]\right)$$

This is easily proved by writing

$$\begin{aligned} Var(y) &= E\big[\,(y - E[y|\mathbf{x}] + E[y|\mathbf{x}] - E[y])^2\,\big] \\ &= E\big[\,(y - E[y|\mathbf{x}])^2\,\big] + E\big[\,(E[y|\mathbf{x}] - E[y])^2\,\big] + 2E\big[\,(y - E[y|\mathbf{x}])\,(E[y|\mathbf{x}] - E[y])\,\big] \end{aligned}$$

The expectation of the cross product on the last line is 0 by Property CE.6, the first term on the second line is $E[Var(y|\mathbf{x})]$ by iterated expectations, and the second term on the second line is $Var(E[y|\mathbf{x}])$ because $E[y]$ is equal to $E[E[y|\mathbf{x}]]$, also by iterated expectations. Property CV.2 can be extended to decompose a conditional variance, and also a conditional covariance; see Property CV.3 and Property CCOV.1 in W. Property CV.4 leads to an important conclusion, as the comment below its statement makes clear.

Conditional Expectation – Simple example. Let's calculate the conditional expectation for a bivariate normal distribution. Let the random vector $\mathbf{w} = (y, x)'$ have a normal distribution with mean vector $\mu = (\mu_y, \mu_x)'$ and covariance matrix

$$\Sigma = \begin{pmatrix} \sigma_y^2 & \sigma_y \sigma_x \rho \\ \sigma_x \sigma_y \rho & \sigma_x^2 \end{pmatrix},$$

where σ_y and σ_x are the standard deviations and ρ is the correlation between y and x. The joint density is

$$f_{Y,X}(y,x) = \frac{1}{2\pi|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{w}-\mu)'\Sigma^{-1}(\mathbf{w}-\mu)\right), \quad -\infty < y, x < \infty.$$

The determinant of the covariance matrix is

$$|\Sigma| = \sigma_y^2\sigma_x^2(1-\rho^2),$$

and the inverse of the covariance matrix is

$$\Sigma^{-1} = \frac{1}{1-\rho^2}\begin{pmatrix} \frac{1}{\sigma_y^2} & -\frac{\rho}{\sigma_y\sigma_x} \\ -\frac{\rho}{\sigma_y\sigma_x} & \frac{1}{\sigma_x^2} \end{pmatrix}.$$

Thus, the joint density can be written as

$$f_{Y,X}(y,x) = \frac{1}{2\pi\sigma_y\sigma_x(1-\rho^2)^{1/2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left[\left(\frac{y-\mu_y}{\sigma_y}\right)^2 - 2\rho\left(\frac{y-\mu_y}{\sigma_y}\right)\left(\frac{x-\mu_x}{\sigma_x}\right) + \left(\frac{x-\mu_x}{\sigma_x}\right)^2\right]\right).$$

The marginal density of x is

$$f_X(x) = \int_{-\infty}^{\infty} f_{Y,X}(y,x)dy = \frac{1}{\sqrt{2\pi}\sigma_x}\exp\left(-\frac{1}{2}\left[\frac{x-\mu_x}{\sigma_x}\right]^2\right),$$

that is, normal with mean μ_x and variance σ_x^2. To perform the integration, complete the square in the exponent and transform via $z = (y-\mu_y)/\sigma_y$. It follows that the conditional density of y, given x, is

$$\begin{aligned} f_{Y|X}(y|x) &= \frac{f_{Y,X}(y,x)}{f_X(x)} \\ &= \frac{1}{\sqrt{2\pi}\sigma_y(1-\rho^2)^{1/2}}\exp\left(-\frac{1}{2\sigma_y^2(1-\rho^2)}\left[y-\mu_y-\rho\frac{\sigma_y}{\sigma_x}(x-\mu_x)\right]^2\right), \end{aligned}$$

that is, normal with (conditional) expectation

$$E[y|x] = \mu_y + \rho\frac{\sigma_y}{\sigma_x}(x-\mu_x)$$

and (conditional) variance

$$Var(y|x) = \sigma_y^2(1-\rho^2).$$

Thus, the conditional expectation of y, given x, is a linear function of x. It is called the regression function, and $\rho\sigma_y/\sigma_x$ is the regression slope. The same linear structure for the conditional expectation occurs for higher dimensional normal distributions. For example, suppose that $(y, \mathbf{x}')$ has a $(K+1)$-dimensional normal distribution. The vector $\mathbf{x}$ is here taken to be $K \times 1$. Then the conditional distribution of y, given $\mathbf{x}$, is normal with mean $\mu_y + \sigma_{y\mathbf{x}}\Sigma_{\mathbf{xx}}^{-1}(\mathbf{x}-\mu_x)$ and variance

$\sigma_y^2 - \sigma_{y\mathbf{x}}\Sigma_{\mathbf{xx}}^{-1}\sigma_{\mathbf{x}y}$, where $\sigma_{y\mathbf{x}}$ is the $1 \times K$ row vector giving the covariances between y and the elements of $\mathbf{x}$ and $\sigma_{\mathbf{x}y}$ is its transpose, and $\Sigma_{\mathbf{xx}}$ is the $K \times K$ covariance matrix of $\mathbf{x}$. Also, the mean vector $\mu_{\mathbf{x}} = E[\mathbf{x}]$ is a column vector. The row vector $\sigma_{y\mathbf{x}}\Sigma_{\mathbf{xx}}^{-1}$ contains the regression slopes (for the regression of y on $\mathbf{x}$).

Important question: What is the implication of this linear structure result?

As we noted previously, the error form of a model specified by a conditional expectation is

$$y = E[y|\mathbf{x}] + u,$$

where u is called the error or disturbance term. As we have noted, from properties of the conditional expectation operator, it follows immediately that

$$E[u|\mathbf{x}] = 0,$$

and this implies $E[u] = 0$. Furthermore, u is uncorrelated with *any* function of $\mathbf{x}$, $g(\mathbf{x})$ (Property CE.6 in W); that is, $E[g(\mathbf{x})u] = 0$, provided $E|g(\mathbf{x})u| < \infty$ and $E|u| < \infty$.

The *partial effect* of x_j on $E[y|\mathbf{x}]$ is defined to be the partial derivative $\partial E[y|\mathbf{x}]/\partial x_j$. The change in $E[y|\mathbf{x}]$ when x_j changes by a small amount and the other variables in $\mathbf{x}$ are held fixed can then be approximated as

$$\Delta E[y|\mathbf{x}] \approx \frac{\partial E[y|\mathbf{x}]}{\partial x_j}\Delta x_j.$$

[Of course, this method can be used to obtain the approximate changes for the conditional expectations in (a)-(h) in appendix]

The (*partial*) elasticity of $E[y|\mathbf{x}]$ with respect to x_j is defined to be

$$\frac{\partial E[y|\mathbf{x}]}{\partial x_j} \times \frac{x_j}{E[y|\mathbf{x}]}.$$

If both $E[y|\mathbf{x}]$ and x_j are positive, this is equal to

$$\frac{\partial \log E[y|\mathbf{x}]}{\partial \log x_j}. \tag{1}$$

Thus, the partial elasticity may be interpreted as the approximate percentage change in $E[y|\mathbf{x}]$ divided by the approximate percentage change in x_j.

Now assume $y > 0$ and consider the model

$$\log y = E[\log y|\mathbf{x}] + v,$$

where v is the error term. It follows that

$$E[y|\mathbf{x}] = 0$$

and $E[v] = 0$. We can rewrite this model as

$$y = \exp\left(E[\log y|\mathbf{x}]\right) \cdot \exp(v).$$

The conditional expectation given $\mathbf{x}$ is

$$E[y|\mathbf{x}] = \exp\left(E[\log y|\mathbf{x}]\right) E\left(\exp(v)|\mathbf{x}\right). \tag{2}$$

If v and $\mathbf{x}$ are independent, then

$$E\left(\exp(v)|\mathbf{x}\right) = E\left(\exp(v)\right). \tag{3}$$

It follows from (2) and (3) that we can write

$$\log E[y|\mathbf{x}] = E[\log y|\mathbf{x}] + \log E[\exp(v)].$$

Therefore, if $x_j > 0$,

$$\frac{\partial \log E[y|\mathbf{x}]}{\partial \log x_j} = \frac{\partial E[\log y|\mathbf{x}]}{\partial \log x_j}.$$

That is, if v and $\mathbf{x}$ are independent, the logarithmic and expectation operations commute for calculation of partial elasticities. Without independence, the operations do not commute. The discussion of this on page 17 of W is a bit incomplete.

In Section 2.2.5 Wooldridge considers partial effects when the structural conditional expectation involves unobserved explanatory variables. Suppose $\mathbf{x}$ is a vector of observed explanatory variables, and q is an unobserved explanatory variable. If the structural conditional expectation is $E[y|\mathbf{x}, q]$, we may want to estimate the partial effect

$$\frac{\partial E[y|\mathbf{x}, q]}{\partial x_j}.$$

However, q is not observed, which poses a problem. We can estimate the effect at the value $q = 0$, but this is of questionable value. The average partial effect of x_j is defined to be

$$E_q\left[\frac{\partial E[y|\mathbf{x}, q]}{\partial x_j}\right],$$

where the expectation is with respect to the distribution of q. Whether one can calculate this in practice is another matter. Wooldridge discusses the use of a proxy variable to tackle the problem posed by an unobserved explanatory variable (more about this later in the course).

Linear Projection. The linear projection and its properties underlie the methodology in this course. Let $y, x_1, ..., x_K$ be random variables with finite second moments. We assume that $E[\mathbf{x}'\mathbf{x}]$ is nonsingular. Define the vector of explanatory variables, $\mathbf{x} = (1, x_1, ..., x_K)$. We now assume that the first element of $\mathbf{x}$ is 1 and that $\mathbf{x}$ is a $1 \times (K+1)$ row vector. (Wooldridge assumes that vector of explanatory variables in regression is a row vector; he assumes all other vectors are column vectors. I am following this convention. Later, unless otherwise stated, I'll assume $\mathbf{x}$ is a $1 \times K$ row vector with first element equal to 1, as Wooldridge does; the assumption being made should be clear from context.) The *linear projection* of y on $\mathbf{x}$ is denoted by

$$L(y|\mathbf{x}) = \beta_0 + \beta_1 x_1 + \cdots + \beta_K x_K = \mathbf{x}\beta,$$

where $\beta = (\beta_0, \beta_1, ..., \beta_K)'$ is a $(K+1) \times 1$ column vector determined by

$$E[\mathbf{x}'\mathbf{x}]\beta = E[\mathbf{x}'y]. \tag{4}$$

Because we assume that $E[\mathbf{x}'\mathbf{x}]$ is nonsingular, β is unique and is given by $\beta = [E[\mathbf{x}'\mathbf{x}]]^{-1}E[\mathbf{x}'y]$.

We refer to the linear projection $\mathbf{x}\beta$ as the *population regression function*. It is also called the (linear) least squares predictor (in the population) of y given $\mathbf{x}$.

Write

$$y = L(y|\mathbf{x}) + u = \mathbf{x}\beta + u = \beta_0 + \beta_1 x_1 + \cdots + \beta_K x_K + u.$$

We emphasize that u, the so-called error term, does not have an existence of its own. It is defined to be the difference between y and $L(y|\mathbf{x})$. (Recall the same discussion about u in the context of conditional expectation.) It satisfies

$$E[u] = 0, \quad Cov(x_j, u) = 0, \quad j = 1, ..., K. \tag{5}$$

This follows because the definition of the projection (4) gives

$$\mathbf{0} = E[\mathbf{x}'\mathbf{x}]\beta - E[\mathbf{x}'y] = E[\mathbf{x}'(\mathbf{x}\beta - y)] = -E[\mathbf{x}'u].$$

Note that $E[u] = 0$ always holds when the first element of $\mathbf{x}$ is 1 (there is an intercept). The conditions on u in (5) are always satisfied when one has a linear projection and an intercept is included (inclusion of the intercept give $E[u] = 0$). See property LP.2 in Appendix 2.A.3 of W. It is common to state that (5) is definitional when u is defined a $y - L(y|\mathbf{x})$ and there is an intercept in the regression. Recall that the conditional expectation $E[y|\mathbf{x}]$ is a projection. However, it is not necessarily a linear projection. Of course, if u is defined from the conditional expectation via $y = E[y|\mathbf{x}] + u$, u satisfies (5) (a special case of property CE.6 in Appendix 2.A.1). Compare CE.6 to (5).

Let's examine (2.39) and (2.40) in W. Changing the notation there, we write $\beta = (\beta_0, \beta_1, ..., \beta_K)' = (\beta_0, \overline{\beta}')'$, $\mathbf{x} = (1, x_1, ..., x_K) = (1, \overline{\mathbf{x}})$. Then $L(y|\mathbf{x}) = \beta_0 + \overline{\mathbf{x}}\overline{\beta}$, and (4) is

$$\begin{pmatrix} 1 & E[\overline{\mathbf{x}}] \\ E[\overline{\mathbf{x}}'] & E[\overline{\mathbf{x}}'\overline{\mathbf{x}}] \end{pmatrix} \begin{pmatrix} \beta_0 \\ \overline{\beta} \end{pmatrix} = \begin{pmatrix} E[y] \\ E[\overline{\mathbf{x}}'y] \end{pmatrix} \tag{6}$$

Rewrite (6) as two equations:

$$\beta_0 + E[\overline{\mathbf{x}}]\overline{\beta} = E[y], \tag{7}$$

$$E[\overline{\mathbf{x}}']\beta_0 + E[\overline{\mathbf{x}}'\overline{\mathbf{x}}]\overline{\beta} = E[\overline{\mathbf{x}}'y]. \tag{8}$$

Multiply (7) on the left by $E[\overline{\mathbf{x}}']$ and subtract the result from (8), yielding

$$\left[E[\overline{\mathbf{x}}'\overline{\mathbf{x}}] - E[\overline{\mathbf{x}}']E[\overline{\mathbf{x}}]\right]\overline{\beta} = E[\overline{\mathbf{x}}'y] - E[\overline{\mathbf{x}}']E[y] \tag{9}$$

Therefore,

$$\overline{\beta} = [Var(\overline{\mathbf{x}})]^{-1} Cov(\overline{\mathbf{x}}, y). \tag{10}$$

The solution for β_0 follows directly from (7), $\beta_0 = E[y] - E[\overline{\mathbf{x}}]\overline{\beta}$. The linear projection is

$$L[y|\mathbf{x}] = \beta_0 + \overline{\mathbf{x}}\overline{\beta} = E[y] - E[\overline{\mathbf{x}}]\overline{\beta} + \overline{\mathbf{x}}\overline{\beta}.$$

This projection is a surface which goes through the point given by the means of the variables, $(E[\overline{\mathbf{x}}], E[y])$.

Property LP.1 in Appendix 2.A.3 of W states that, if the conditional expectation is linear, it is equal to the linear projection. Let's consider the general form of this result. Suppose

$$E[y|\mathbf{x}] = \beta_1 g_1(\mathbf{x}) + \cdots + \beta_J g_J(\mathbf{x}) = \mathbf{g}(\mathbf{x})\beta,$$

where $\mathbf{g}(\mathbf{x}) = (g_1(\mathbf{x}), ..., g_J(\mathbf{x}))$ is a $1 \times J$ row vector. By CE.6,

$$\mathbf{0} = E\left[\mathbf{g}(\mathbf{x})'u\right] = E\left[\mathbf{g}(\mathbf{x})'\left[y - E[y|\mathbf{x}]\right]\right] = E\left[\mathbf{g}(\mathbf{x})'\left[y - \mathbf{g}(\mathbf{x})\beta\right]\right].$$

Therefore,

$$E[\mathbf{g}(\mathbf{x})'\mathbf{g}(\mathbf{x})\beta] = E[\mathbf{g}(\mathbf{x})'y]$$

That is,

$$L(y|\mathbf{g}(\mathbf{x})) = \mathbf{g}(\mathbf{x})\beta = E[y|\mathbf{x}].$$

Next, consider the linear projection of the conditional expectation $E[y|\mathbf{x}]$. This is equal to the linear projection $\mathbf{x}\beta$ itself. That is, $L(E[y|\mathbf{x}]|\mathbf{x}) = L(y|\mathbf{x})$. See Property LP.5 in Appendix 2.A.3 of W [also (2.45)], which is slightly more general; in addition, see Problem 2.9 in W. To prove this, write

$$E[\mathbf{x}'\mathbf{x}]\beta = E[\mathbf{x}'y] = E[\mathbf{x}'E[y|\mathbf{x}]], \tag{11}$$

where the second equality follows from iterated expectations. Expression (11) implies that, in performing linear regression, one obtains the correct projection if one works with grouped y values (averaged y values). While this is true, working with grouped y values will not yield the correct standard errors for the original microdata.

The linear projection is in fact the minimum mean square linear predictor, or the least squares linear predictor, of y, given $\mathbf{x}$. This follows because β is the choice of $\mathbf{b}$ which minimizes $E[(y-\mathbf{x}\mathbf{b})^2]$. This assertion is property LP.6 in Appendix 2.A.3 of W.

Appendix 2.A.3 of W gives properties of linear projections. I've already discussed LP.1, LP.2, LP.5, and LP.6. Property LP.3 says a linear projection is a linear operator; it is analogous to CE.1.

Property LP.4 describes iterated projections. It is the analog of part (2) of CE.3. Let's examine a consequence of it. Let y be a response variable, and suppose the explanatory variables are given by $(\mathbf{x},\mathbf{z})$. Consider the two linear projections

$$L(y|\mathbf{x},\mathbf{z}) = \mathbf{x}\beta + \mathbf{z}\gamma, \quad L(y|\mathbf{x}) = \mathbf{x}\delta.$$

The first is a long regression and the second is a short regression. In error from, the first of these is

$$y = \mathbf{x}\beta + \mathbf{z}\gamma + u,$$

where $E[\mathbf{x}'u] = E[\mathbf{z}'u] = \mathbf{0}$. LP.4 states that

$$L(y|\mathbf{x}) = L(L(y|\mathbf{x},\mathbf{z})|\mathbf{x}). \tag{12}$$

Moreover,

$$\begin{aligned} L(y|\mathbf{x}) &= L(\mathbf{x}\beta + \mathbf{z}\gamma|\mathbf{x}) \quad \text{by LP.4} \\ &= L(\mathbf{x}|\mathbf{x})\beta + L(\mathbf{z}|\mathbf{x})\gamma \quad \text{by LP.3.} \end{aligned} \tag{13}$$

But,

$$L(\mathbf{x}|\mathbf{x}) = \mathbf{x}, \quad L(\mathbf{z}|\mathbf{x}) = \mathbf{x}\left(E[\mathbf{x}'\mathbf{x}]\right)^{-1} E[\mathbf{x}'\mathbf{z}] = \mathbf{x}\Pi,$$

where

$$\Pi = \left(E[\mathbf{x}'\mathbf{x}]\right)^{-1} E[\mathbf{x}'\mathbf{z}].$$

Therefore, (12) is

$$\mathbf{x}\delta = \mathbf{x}\beta + \mathbf{x}\Pi\gamma = \mathbf{x}(\beta + \Pi\gamma).$$

That is, $\delta = \beta + \Pi\gamma$, which is the *omitted variables bias* formula (in the population). If you should perform the long regression with explanatory variables $\mathbf{x}$ and $\mathbf{z}$, but you include only the explanatory variables $\mathbf{x}$ in your regression (that is, you perform the short regression), $\Pi\gamma$ is the bias you experience in estimating the $\mathbf{x}$ coefficients. See Section 2.3 in W.

Property LP.7 is interesting. It is useful and it provides understanding of what a partial slope in a regression represents. The covariate vector is partitioned into two parts as $(\mathbf{x}, \mathbf{z})$, and the linear projection of y is denoted

$$L(y|\mathbf{x}, \mathbf{z}) = \mathbf{x}\beta + \mathbf{z}\gamma, \tag{14}$$

as above. Form the regressions of the elements of $\mathbf{x}$ on $\mathbf{z}$ and denote the (vector of) residuals from this operation by

$$\mathbf{r} = \mathbf{x} - L(\mathbf{x}|\mathbf{z}).$$

Form the regression of y on $\mathbf{z}$ and denote the residual from this operation by

$$v = y - L(y|\mathbf{z}).$$

Now form the regression of v on $\mathbf{r}$ – residual on residuals (the correlation with $\mathbf{z}$ has been stripped out of both y and $\mathbf{x}$). This operation yields

$$L(v|\mathbf{r}) = \mathbf{r}\beta. \tag{15}$$

The point is that the parameter vector β in (15) is the same as the β in (14). In addition to (15), it is true that $L(y|\mathbf{r}) = \mathbf{r}\beta$; that is, one does not have to use regression to remove the linear effect of $\mathbf{z}$ from y. A proof is in W.

Let's prove a special case of LP.7. We'll change some notation here for convenience. Consider the vector of explanatory variables

$$\mathbf{x} = (1, x_1, ..., x_K),$$

and define

$$\mathbf{x}_{(-j)} = (1, x_1, ..., x_{j-1}, x_{j+1}, ..., x_K).$$

Further, define

$$\hat{x}_j = x_j - L(x_j|\mathbf{x}_{(-j)}),$$

the residual from the regression of x_j on $\mathbf{x}_{(-j)}$.

The linear projection of y on $\mathbf{x}$ is $L(y|\mathbf{x})$, and

$$y = L(y|\mathbf{x}) + u = \beta_0 + \beta_1 x_1 + \cdots + \beta_j x_j + \cdots + \beta_K x_K + u.$$

LP.7 implies

$$\beta_j = \frac{E[y\hat{x}_j]}{E[\hat{x}_j^2]}.$$

This gives an interpretation of a partial slope in a multiple regression as a marginal slope in a specific simple regression. The proof is straightforward. It follows from

$$\begin{aligned}
E[y\hat{x}_j] &= E\Big[\,(\beta_0 + \beta_1 x_1 + \cdots + \beta_{j-1}x_{j-1} + \beta_j x_j + \beta_{j+1}x_{j+1} + \cdots + \beta_K x_K + u)\,\hat{x}_j\Big] \\
&= E\Big[\,(\beta_0 + \beta_1 x_1 + \cdots + \beta_{j-1}x_{j-1} + \beta_j x_j + \beta_{j+1}x_{j+1} + \cdots + \beta_K x_K)\,\hat{x}_j\Big] \quad \text{by LP.2} \\
&= \beta_j E[x_j\hat{x}_j] \quad \text{again by LP.2} \\
&= \beta_j E[\hat{x}_j^2] \quad \text{again by LP.2}
\end{aligned}$$

Appendix

Interpretation of Partial Effects with Presence of log Transformation. Before we turn to a careful discussion of conditional expectations and linear projections, let's consider the interpretation of partial effects in some models. Most of these discussed here employ log transformations of the response and/or explanatory variables. See the examples in Chapters 1 and 2 of W, and especially the discussion in Section 2.2.2. Suppose y is a response and $\mathbf{x} = (x_1, ..., x_K)$ is a row vector of explanatory variables. Consider the conditional expectation $E[y|x]$ (we assume that $E[|y|] < \infty$). We can write

$$y = E[y|x] + u. \tag{16}$$

Here we note that u in (16) is not a variable with an existence of its own. It is defined as the difference between y and $E[y|x]$. From properties of the conditional expectation (discussed below), we conclude that

$$E[u|x] = 0. \tag{17}$$

In some cases we will write, alternatively,

$$\log y = E[\log y|\mathbf{x}] + u. \tag{1'}$$

We view the conditional expectation as the quantity we want to study. The expression (16) is the error from of the model of the response y, and (1') is the error form when the response is $\log y$.

Now consider the change in $E[y|\mathbf{x}]$ which results when $\mathbf{x}$ is changed, for some common linear forms of the conditional expectation which we will encounter. Throughout this discussion I am assuming that the elements of $\mathbf{x}$ are continuous variables, and that the logarithm to the base e is being used. If an element of $\mathbf{x}$ is discrete, on the other hand, we will simply compare the values of y at the different values of this element. For the first six examples which follow, $\mathbf{x}$ will have a single element, but this is no restriction.

(a) $E[y|x] = \beta_0 + \beta_1 x$. If x changes to $x + \delta$, $E[y|x]$ changes additively by the amount $\beta_1\delta$.

(b) $E[y|x] = \beta_0 + \beta_1 \log x$. If x changes to $x + \delta$, the change in $E[y|x]$ is additively $\beta_1 \log(1 + \delta/x)$. From the Taylor expansion of the log function, this change is approximately $\beta_1\delta/x$. This approximation is good if δ is small. If the change is multiplicative, that is, if x changes to cx for some positive c, then $E[y|x]$ changes additively by the amount $\beta_1 \log c$.

(c) $E[\log y|x] = \beta_0 + \beta_1 x$. From (1') we have $\log y = \beta_0 + \beta_1 x + u$. Then,

$$y = \exp(\beta_0)\exp(\beta_1 x)\exp(u), \text{ so that } E[y|x] = \exp(\beta_0)\exp(\beta_1 x)E[\exp(u)|x].$$

Assume that u and x are *independent*. Then $E[\exp(u)|x] = E[\exp(u)]$, and thus

$$E[y|x] = \exp(\beta_0)\exp(\beta_1 x)E[\exp(u)].$$

Suppose x changes to $x + \delta$. Then,

$$\frac{E[y|x+\delta]}{E[y|x]} = \frac{\exp(\beta_1(x+\delta))}{\exp(\beta_1 x)} = \exp(\beta_1\delta).$$

Thus, if x changes to $x+\delta$, the change in $E[y|x]$ is $100[\exp(\beta_1\delta)-1]$ per cent. From Taylor expansion of the exponential function, we see that the approximate change in $E[y|x]$ is $100\beta_1\delta$ per cent, valid for small δ.

(d) $E[\log y|x] = \beta_0 + \beta_1 \log x$. We proceed as in (c), assuming that u and x are independent. Write

$$E[y|x] = \exp(\beta_0)x^{\beta_1}E[\exp(u)].$$

If x changes to $x+\delta$,

$$\frac{E[y|x+\delta]}{E[y|x]} = \left(1+\frac{\delta}{x}\right)^{\beta_1}.$$

Thus, the change in $E[y|x]$ is $100[(1+\delta/x)^{\beta_1}-1]$ per cent. For small δ, the approximate change is $\beta_1(100\delta/x)$, that is, β_1 times the percentage change in x. To see this, use Newton's binomial formula (I'll discuss it in class).

(e) $E[y|x] = \beta_0 + \beta_1\log x + \beta_2(\log x)^2$. If x changes to $x+\delta$, the (additive) change in $E[y|x]$ is

$$\begin{aligned}
\beta_1 \log\left(1+\frac{\delta}{x}\right) + \beta_2\left[(\log(x+\delta))^2 - (\log x)^2\right] &= \beta_1\log\left(1+\frac{\delta}{x}\right) + \beta_2\log\left(1+\frac{\delta}{x}\right)[\log(x+\delta)+\log(x)] \\
&= \beta_1\log\left(1+\frac{\delta}{x}\right) + 2\beta_2\log\left(1+\frac{\delta}{x}\right)\log\left(x\left(1+\frac{\delta}{x}\right)^{1/2}\right) \\
&= \left(\beta_1 + 2\beta_2\log\left(x\left(1+\frac{\delta}{x}\right)^{1/2}\right)\right)\log(\left(1+\frac{\delta}{x}\right).
\end{aligned}$$

For small δ, this change is approximately (use Newton's binomial formula)

$$[\beta_1 + 2\beta_2\log(x+\delta/2)]\frac{\delta}{x}.$$

Thus, in this case, the change depends on the initial value of x. Now suppose the change in x is multiplicative, from x to cx, for some positive c. Then the (additive) change in $E[y|x]$ is a change by the amount

$$\beta_1\log c + \beta_2(\log c + 2\log x)\log c.$$

(f) $E[\log y|x] = \beta_0 + \beta_1 x + \beta_2 x^2$. Proceed as in (c), assuming u and x are independent. This gives

$$\frac{E[y|x+\delta]}{E[y|x]} = \frac{\exp\left(\beta_1(x+\delta) + \beta_2(x+\delta)^2\right)}{\exp\left(\beta_1 x + \beta_2 x^2\right)}.$$

Thus, the change in $E[y|x]$ is

$$100\{\exp[\beta_1 + 2\beta_2(x + \delta/2))\delta] - 1\} \text{ per cent.}$$

If δ is small, this percentage change is approximately $100(\beta_1 + 2\beta_2 x)\delta$.

These examples illustrate interpretation of models when there is a single explanatory variable. Typically, we need to deal with multiple explanatory variables. If the conditional expectation is a function of K variables, we can proceed as in the above examples. Of course, we can and often do consider the change in $E[y|x]$ when more than one of the explanatory variables change. Here are two simple examples:

(g) $E[y|x_1, x_2] = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_2^2 + \beta_4 x_1 x_2$.

(h) $E[\log y|x_1, x_2] = \beta_0 + \beta_1 x_1 + \beta_2 \log x_2$.

(Optional reading) Experiments, Observational Studies, and Causal Inference. As noted, a prominent goal of econometric studies is to establish causal inference. Often, however, there are obstacles to drawing such inference. If data are collected in a controlled experiment with random assignment of treatments to subjects, and if the experiment proceeds according to plan, one can usually make causal inference. However, if there is noncompliance by some of the subjects, a common occurrence, drawing proper inference can become difficult. And other complications can result also.

Making inference with observational studies, where treatments are not randomly assigned to subjects, is more difficult. One approach is to divide the data into subsets where one is comparing like to like. For example, suppose one wants to consider the incidence of heart disease among smokers vs. nonsmokers. One can subdivide the data by holding constant the variables gender, age, economic status, race, etc., and make the comparison within each subset. The drawback of this approach is that the subsets can end up containing very few data points, resulting in little statistical power. Also, the subsetting may have failed to take account of a key variable, resulting in confounding.

In addition to randomized controlled experiments and observational studies, there are natural experiments which can be exploited. Later in the course we will encounter an example of use of a natural experiment.

The book by D. A. Freedman (*Statistical Models, Theory and Practice*, rev. ed., Cambridge University Press, 2009) discusses the HIP experiment. This was the first large-scale randomized controlled trial to assess the effectiveness of mammography in detecting breast cancer early enough to reduce the death rate from the disease. The trial took place in the 1960s. At the time the Health Insurance Plan (HIP) of Greater New York had 700,000 enrollees. From these, 62,000 women aged 40–64 were selected and randomly assigned to treatment and control. Those in the treatment group were invited to four rounds of screening annually. Each screening visit involved mammography and a clinical examination. Those assigned to the control group received usual health care.

Follow-up continued for 18 years. Freedman gives the following table showing results from the first five years of follow-up. The table shows the number of deaths and the death rate per 1000 women during the five-year follow-up period.

		Group size	Breast cancer deaths		All other deaths	
			Number	Rate	Number	Rate
Treatment						
	Screened	20,200	23	1.1	428	21
	Refused	10,800	16	1.5	409	38
	Total	31,000	39	1.3	837	27
Control						
	Total	31,000	63	2.0	879	28

The table shows that about 1/3 of the women assigned to treatment declined to take part in the screening. It is not appropriate to compare the women who were screened to those who refused to participate. Such a comparison introduces bias. Those who participated tended to be women who were wealthier and more educated, and those who refused tended to be poorer and less well educated. That is, social status is a confounding factor in such a comparison. The last column in the table confirms this reasoning. The death rate from causes other than breast cancer is almost twice as great for those who refused as for those who were screened. Wealthier, more educated women are less vulnerable to other diseases. However, they are in fact more vulnerable to breast cancer than poor, less well educated women. The latter tend to have more children, and pregnancy protects against breast cancer, because it decreases the number of lifetime menstrual cycles. Thus, those who were screened differ in a material way from those who declined to be screened. A comparison of these two groups will result in biased estimation.

If we compare the death rate for those who were screened to the rate for those who refused, we are conducting analysis by *treatment received*, and, as noted, this results in bias. A valid comparison for this experiment is *intention-to-treat analysis*, which contrasts those in the total treatment group to those in the control group. These two groups have equal numbers of subjects, and there were

63–39=24 fewer breast cancer deaths in the total treatment group. Put another way, the ratio 39/63=0.62 corresponds to a 38 per cent decrease in deaths. Improvement persisted over the entire 18 years of followup.

A large randomized controlled experiment in Sweden confirmed these HIP findings, as did several other subsequent studies. These results led to the wide acceptance of mammography as a screening tool for breast cancer.

(Optional reading) Selection Bias. A paper by M. Baiocchi, D. S. Small, S. Lorch, and P. R. Rosenbaum (2010, Building a stronger instrument in an observational study of perinatal care for premature infants, *J. Amer. Statist. Assoc.*, Vol. 105, pp. 1285–1296) addresses survival of premature infants at hospitals.

The authors give the following background:

> The American Academy of Pediatrics recognizes six levels of neonatal intensive care units (NICUs) of increasing technical expertise and capability: 1, 2, 3A, 3B, 3C, 3D, and regional centers, 4. The term "regionalization of care" refers to a policy that suggests or requires that high-risk mothers deliver at hospitals with greater capabilities. In other words, within a region, mothers are to be sorted into hospitals of varied capability based on the risks faced by the newborn, rather than on haphazard circumstances, such as affiliation or proximity. Regionalized perinatal systems were developed in the 1970s, when NICUs began to save infants with birth weight < 1500 g. In the 1990s, however, NICU services began to diffuse from regional centers to community hospitals. Regionalization might reduce infant mortality by bringing together the sickest babies and the most capable hospitals; however, regionalization might not reduce infant mortality because the sorting by risk might be too inaccurate to affect health, or the capabilities of high-level NICUs might fail to deliver better outcomes.
>
> In the current paper, we focus on whether delivering high risk infants at more capable NICUs reduces mortality... More precisely, if a high-risk mother delivers at a less capable hospital, is her baby at greater risk of death? In a highly abstract world remote from the world that we inhabit a randomized experiment could settle that question, with high-risk mothers assigned at random to hospitals of varied capabilities. In the world that we actually do inhabit, in which medical decisions are happily constrained by considerations of sound judgement, ethics, and patient preferences, such an experiment is not possible. We need to make some reasonable sense of the data that we can obtain. There is a basic difficulty, however, that arises in many contexts in which the most intense, and capable care is given to the sickest patients. If regionalization succeeded in sorting mothers by risk, then the highest-risk mothers would deliver at the most-capable

hospitals. The mortality rates at the more-capable hospitals might be higher, not lower, than those at the less-capable hospitals because their patient populations were sicker, even if the more-capable hospitals were saving lives. A naive comparison of mortality rate by level of NICU would do little or nothing to clarify whether regionalization is or is not effective, because it would not estimate the effect on mortality of delivery at a more-capable hospital.

The context of the authors' study is thus an observational study, rather than a designed experiment. The main problem one faces in an observational study is *selection bias.* In the present setting, this occurs because the patients at greater risk are sent to the more capable hospitals. That is, babies delivered at high level NICUs tend to differ from babies delivered at low level NICUs.

To address causality here, we consider a *potential outcomes* framework. This concept was apparently first proposed in 1923 by Jerzy Neyman. Development of the framework to study causation is prominently due to Donald Rubin and his colleagues, including Paul Rosenbaum.

Let

$$\begin{aligned} D_i &= 1 \quad \text{if child } i \text{ is delivered at a high level NICU} \\ &= 0 \quad \text{otherwise.} \end{aligned}$$

This is the observed treatment. The observed outcome is also binary:

$$\begin{aligned} Y_i &= 1 \quad \text{if child } i \text{ dies,} \\ &= 0 \quad \text{if child } i \text{ does not die.} \end{aligned}$$

(Here I am using capital letters for the random variables)

Next, let's specify the potential outcomes:

Y_{1i} is the outcome child i would have if delivered at a high level NICU (if $D_i = 1$),

Y_{0i} is the outcome child i would have if delivered elsewhere (if $D_i = 0$).

For child i the causal effect of being delivered at a high level NICU is therefore $Y_{1i} - Y_{0i}$. However, we only observe one of these two potential outcomes. The average causal effect is

$$E[Y_{1i} - Y_{0i}] = E[Y_{1i}] - E[Y_{0i}].$$

Note that this depends only on the marginal distributions of Y_{1i} and Y_{0i}, not on their joint distribution.

We can write for the outcome

$$
\begin{aligned}
Y_i &= Y_{1i} \text{ if } D_i = 1, \\
&= Y_{0i} \text{ if } D_i = 0, \\
&= Y_{0i} + (Y_{i1} - Y_{0i})D_i.
\end{aligned}
$$

The naive comparison of average outcomes based on NICU status is then

$$
\begin{aligned}
E[Y_i|D_i = 1] - E[Y_i|D_i = 0] &= E[Y_{1i}|D_i = 1] - E[Y_{0i}|D_i = 0] \\
&= E[Y_{1i}|D_i = 1] - E[Y_{0i}|D_i = 1] + E[Y_{0i}|D_i = 1] - E[Y_{0i}|D_i = 0].
\end{aligned}
\tag{18}
$$

The difference on the second line of (18),

$$
E[Y_{1i}|D_i = 1] - E[Y_{0i}|D_i = 1], \tag{19}
$$

is called the average treatment effect on the treated. In this case, it is the average causal effect of being delivered at a high level NICU for those who were delivered at a high level NICU.

However, (19) is contaminated by the third line of (18),

$$
E[Y_{0i}|D_i = 1] - E[Y_{0i}|D_i = 0], \tag{20}
$$

which is the selection bias term. In this example, it is the difference in the average Y_{0i} between those who were delivered at a high level NICU and those who were delivered elsewhere. For this example, the selection bias term is positive, because babies delivered at a high level NICU tend to be less viable than those delivered elsewhere – hence tend to have a higher mortality rate.

At the end of Section 2.1 in MHE, Angrist and Pischke write:

> The goal of most empirical economic research is to overcome selection bias, and therefore to say something about the causal effect of a variable like D_i.

One way to eliminate selection bias is to use a randomized experiment. That is, assign the values of D_i at random. Such random assignment makes D_i independent of the potential outcomes. Then (20) is 0, because $E[Y_{0i}|D_i = 1]$ and $E[Y_{0i}|D_i = 0]$ are equal. Or, we can understand this absence of selection bias by writing

$$
\begin{aligned}
E[Y_i|D_i = 1] - E[Y_i|D_i = 0] &= E[Y_{1i}|D_i = 1] - E[Y_{0i}|D_i = 0] \\
&= E[Y_{1i}|D_i = 1] - E[Y_{0i}|D_i = 1] \\
&= E[Y_{1i} - Y_{0i}|D_i = 1] \\
&= E[Y_{1i} - Y_{0i}],
\end{aligned}
$$

where the second and last equalities follow from independence.

Angrist and Pischke mention that, although randomization eliminates selection bias, randomized experiments are not necessarily free of problems. See their discussion in Section 2.2 of MHE for this and other examples. Also, see P. Frumento, F. Mealli, B. Pacini, and D. B. Rubin (2012, Evaluating the effect of training on wages in the presence of noncompliance, nonemployment, and missing outcome data, *J. Amer. Statist. Assoc.*, Vol. 107, pp. 450–466). We also saw a problem with a randomized experiment in discussion of the HIP experiment.

We can also treat the case where D_i assumes more than two values, or even a continuum of values.

An interesting review paper which addresses instrumental variables and, in doing so, contrasts the approaches of the econometric and statistical literatures, is by G.W. Imbens (2014, Instrumental variables: An econometrician's perspective, *Statist. Sci.*, Vol. 29, pp. 323–358). The paper treats some of the topics I've touched on briefly in these notes.

The August 2020 issue of Statistical Science is a special issue on causal inference. The editors are Jason Roy and Dylan Small. I've posted the editorial and two of the articles (without their added discussions) on Canvas.